



DINIS FERREIRA

Electrical & Computer Engineer |
Robotics, AI & Control Specialist
EU Citizen

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Address

Felgueiras, Portugal
(Willing to relocate)

Digital Portfolio

Experience

MSc Thesis Researcher

University of Coimbra & OnControl

Oct 2025 – Sep 2026

Developed a Master's dissertation titled "Detection of Faults in Electric Motors using Acceleration Sensors" in collaboration with OnControl. The research focused on advanced signal processing and vibration analysis to implement predictive maintenance models and AI-driven fault detection in industrial environments.

Research Fellow (SOMIPS Project)

University of Coimbra & OnControl

May 2026 – Sep 2026

Research fellow within the SOMIPS project, a co-funded EU initiative (FEDER/Centro 2030) in cooperation with OnControl. The work involved developing advanced control solutions and predictive maintenance strategies to optimize the performance and energy efficiency of biomass incineration plants.

Robotics & Programming Instructor

RoboThink

2024–2025

Planned and delivered practical robotics and programming lessons for students aged 6 to 14 using specialized robotics kits and educational software. The role focused on fostering logical reasoning and creativity through the development of weekly mechatronic projects and hands-on technical challenges

IT Tutor for Seniors

ASSP

2024–2026

Conducted IT courses for seniors aimed at promoting digital inclusion and technological literacy. Responsibilities included teaching basic computing, internet navigation, and online safety while adapting pedagogical methods to meet the diverse learning paces and needs of the participants.

Technical Draftsman

ToogoCasa — Projetos e Construção

2022–2025

Performed high-precision technical drawing using AutoCAD for architectural and engineering projects. The work involved the interpretation and execution of technical plans, ensuring geometric accuracy for various construction and specialty projects.

Education

BSc in Electrical and Computer Engineering

University of Coimbra

This 180 ECTS program provided a strong technical foundation in automation, control systems, and signal processing. The degree focused on applying information technologies to industrial maintenance and optimization tasks

MSc in Electrical and Computer Engineering

University of Coimbra

MSc in Electrical and Computer Engineering (Level 7 EQF) specialised in Robotics, Control, and Artificial Intelligence. This advanced cycle of studies builds upon my undergraduate foundation, focusing on my current research in sensor-based fault detection and predictive maintenance for industrial reliability.

About Me

Electrical and Computer Engineering graduate (M.Sc.) from the University of Coimbra, specializing in Robotics, Control, and AI. Developed a master's thesis in Predictive Maintenance focused on signal processing and fault detection for industrial motors, in collaboration with OnControl. My research includes the SOMIPS project, aimed at developing advanced control and maintenance solutions for industrial process optimisation.

Beyond my technical background in AutoCAD and industrial automation, I have a strong foundation in communication and adaptability gained from teaching robotics to children and IT to seniors. I am a native Portuguese speaker, fluent in English, and currently learning German.

Skills

- Robotics & Control Systems
- Signal Processing
- CAD/CAM & Prototyping
- Machine Learning
- Python
- Matlab & Simulink
- Ladder Logic
- Vibration Analysis
- Computer Vision

Languages

English

Portuguese

German (Currently Learning)